

Can we detect punches and kicks in Thai-Boxing videos using AI methods



TECHNISCHE
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April 20, 2025

1 Abstract

The recognition and analysis of movements and techniques is an important aspect of evaluating the performance in sports. Especially in Thai boxing, even small changes in technique can greatly impact the performance and effectiveness. This project introduces a video-based system to detect and classify Thai boxing techniques by using AI-driven methods. We developed a system, where MediaPipe is used for feature extraction, Dynamic Time Warping (DTW) for temporal sequence comparison and a Random Forest Classifier to categorize movements into 14 distinct Thai boxing techniques.

Our primary objective was to determine the possibility of detecting punches and kicks from video footage with the focus on Zero/Few-Shot Learning and the possible integration of velocity and angles to improve accuracy. We created our own dataset by collaborating with our university Thai boxing club, where we captured multi-angle recordings of various techniques. Despite initial challenges with object detection models, which struggles with the dynamic and continuous nature of strikes, we transitioned to a temporal sequence analysis through DTW and a Random Forest Model based on keypoints. Our System demonstrated promising results, in which leveraging keypoint features, combined with velocity and joint angle data, enhances classification performance. However, limitations such as occluded keypoints, varying camera perspectives and MediaPipe's single-person tracking turned out to be challenging. Nevertheless, through testing and evaluating, we found out, that while DTW struggled with real-world variability, the Random Forest model offered greater flexibility and accuracy.

Our findings highlight the potential of AI-driven movement classification for sports applications, while also draw attention to the importance of diverse datasets and adaptive models for variability. Future work will focus on expanding the dataset, adding multi-angle viewpoints and exploring deep learning techniques to further improve performance and generalization.

2 Introduction

In recent years, performance analysis in sports has undergone a big change, through advancements in technology and the increasing integration of computer science methodologies. Manual analysis presents challenges, including its time-consuming nature and its vulnerability to human error and bias. These limitations call for better and automated solutions capable of recognizing sport movements and techniques and thus delivering more efficient and accurate assessments to the performances. In particular Artificial Intelligence (AI), especially through machine learning methods, offers a promising approach to recognize movement patterns and techniques better [1].

We are five computer science students interested in sports and wanted to apply modern technology to a real athletic discipline. Our project focuses on Thai boxing, a sport that requires precision, speed, and proper technique. In Thai-Boxing, even small deviations in form can impact performance, effectiveness and injury risk.

The primary goal of this project is to develop a system to detect Thai boxing movements. Additionally, we implement visualization tools to enhance comprehensibility. The system could be beneficial for athletes, trainers and enthusiasts looking to enhance their techniques in this sport.

We focus our work on the following Research Questions:

- **RQ1: Can we detect punches and kicks in Thai-Boxing videos using AI methods?**
 - **Sub-RQ 1.1: Can we do Zero/Few-shot Learning to detect fight movement in Thai-Boxing?**
 - **Sub-RQ 1.2: Can we extract the movements of one Person with a punching bag?**

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- **Sub-RQ 1.3:** Can we detect movements of two people in a fighting scenario?
 - **Sub-RQ 1.4:** Does leveraging explicit information about velocity and angles of body parts enhance the system's performance?
 - **Sub-RQ 1.5:** Can we detect all techniques equally well from one perspective?

This report is structured as follows:

- Section 3: Project Overview - Describes the project topic, key requirements, and technologies used in our implementation
- Section 4: Development Process – Covers our planning phase, design decisions, implementation steps, and challenges encountered during development.
- Section 5: Teamwork and Contribution Evaluation - Details how our team collaborated, the task distribution, individual contributions and an consideration of whether every team member should receive the same grade.
- Section 6: Testing and Evaluation - Discussed our testing methodology, results of functional and performance tests and key observations.
- Section 7: Conclusion and Reflection - Summarizes our accomplishments, lessons learned and potential future improvements.
- Section 8: References - List of all research papers, resources and relevant materials
- Section 9: Appendices - Provide additional materials, including source code snippets, diagrams and detailed specifications.

3 Project Overview

The project aims to develop a machine learning system to detect 14 different Thai boxing techniques in videos with the focus on sequential motion patterns.

The system is designed with the following key requirements:

- **Accurate Motion Detection:** The system must be capable of detecting and analyzing various kickboxing movements, including punches and kicks.
- **Video Processing and Feature Extraction:** The system should automatically process video input to extract key body landmarks using MediaPipe Pose and calculate extra features, like velocity and angle.
- **Movement Classification:** The system should be able to classify different types of strikes and kicks.
- **Visualization Tools:** The system should include visualization methods for showcasing the predicted techniques.

To implement this project, we utilized the following core technologies and tools:

- **MediaPipe Pose:** This framework was used for feature extraction from video footage. MediaPipe Pose detects and tracks 33 body landmarks in real time, allowing for accurate representation of the athlete's body movements. [2]
- **OpenCV:** OpenCV was utilized for video processing and manipulation, such as frame extraction, handling video input, and output visualization. It also facilitated the integration of the video footage with the motion analysis algorithm. [3]
- **Dynamic Time Warping (DTW):** DTW was used to compare and classify movement sequences by measuring the similarity between time-series data.[4]
- **RandomForest:** RandomForest was used for movement classification based on features extracted from the video data.[5]
- **Real-Time Performance:** While not a primary goal, the system should be capable of near-real-time processing for practical use in training environments. This is only feasible with Random Forest and not with DTW.
- **Roboflow:** Roboflow was used to label and annotate the movement data efficiently, creating a dataset that could be used for training and testing the motion recognition system.[6]
- **YOLO (You Only Look Once):** YOLO was applied for real-time object detection in video footage, aiding in tracking specific athletes or movements during the analysis process.[7]
- **Discord:** We collaborated through Discord, where we communicated, and coordinated the development process.
- **Git:** We used Git for version control of our code.

3.1 Dataset creation

We met the Student Thai-Boxing Club of TU Darmstadt in order to create a high-quality dataset of clean executions of various fighting techniques. We filmed 4 students with different level of professionalism in Thai-Boxing from three camera perspectives. We asked every student to perform 14 techniques with five to seven repetitions for each technique in 10 to 30 seconds long video sequences.

3.2 Object detection

Our initial attempt at detecting thai boxing strikes (kicks and punches) on video was to utilize object detection models to inspect single frames during impact time i.e., when a person punches a punching bag. To accomplish this, we utilized Roboflow [6], a collaborative dataset-building and preprocessing environment. We manually curated a dataset by extracting key frames from training videos where the strike made contact with the bag. This dataset was then trained using YOLOv11 [7], a state-of-the-art object detection model, to identify the human subject and classify the action as either a punch or a kick.

However, our results revealed significant limitations in relying solely on object detection for this task. One of the important insights was that a strike is not a single frame event but a dynamic sequence of movements. Object detection, which operates on frames in isolation, could not see the temporal evolution required to identify techniques like punches or kicks. A punch, for example, consists of leading movements like shoulder rotation and foot positioning that occur over a number of frames. By taking into account only the impact frame, the model did not have contextual information to distinguish between types of strikes or even between intentional strikes and unintentional movements.

We also encountered problems related to environmental and positional noise. The model could not distinguish the subject from complex backgrounds, particularly in scenes with moving objects or varying light. Moreover, the detection logic was ambiguous during rapid movement. For instance, in a high kick, the leg returns after impact, and it would sometimes trigger a false low kick detection.

3.3 Dynamic Time Warping

To address the frame-based object detection drawback, we switched to temporal sequence analysis using Dynamic Time Warping (DTW) [4]. This method focuses on capturing the entire motion trajectory of strikes, rather than isolated frames, by comparing input video sequences to pre-recorded reference templates of thai boxing techniques. For each strike type (e.g., hooks, kicks), we created reference sequences from expert performances. We stored every reference as the complete time series of extracted keypoint information, this includes x, y, and z coordinate as well as visibility per keypoint. These sequences, saved as time series of landmark data, are the best performance of each strike to use as a comparison basis. DTW is highly efficient, requiring only the storage of reference sequences [4]. Input videos are broken down into overlapping 35-frame segments using a sliding window approach in order to localize possible strikes. We tried manually and found that the best value for the window size was 35 frames. Similarity is measured by the Euclidean distance between aligned landmark trajectories, with lower distances indicating higher matches. If the distance is below a certain threshold, we accept it as a punch, otherwise not [4]. Hyperparameters such as window size and thresholds were determined through manual evaluation.

Figure 1 shows the comparison of a reference sequence (blue dashed line) and a test sequence (orange dashed line) for the movement of the landmark 16 along the X-coordinate. The gray lines connect the temporally assigned points of both sequences according to DTW pathfinding. A closer path indicates a high degree of agreement, while larger deviations indicate differences in the execution of the movement. This illustrates the flexibility of DTW in aligning movement patterns despite temporal variability.

3.4 Random Forest Classifier

As an alternative sequence comparison method, we experimented with a machine learning classification method based on Random Forests (RF) [5]. This method analyzes both body pose and movement dynamics within individual frames. We extract three types of features from MediaPipe's 33 pose landmarks [2]: **Joint angles** for the elbows (calculated through shoulder-elbow-wrist) and the knees (calculated through hip-knee-ankle) through 3D vector geometry, **linear wrist/ankle velocities** calculated from position differences between frames, **angle velocities** measuring the rate at which joints rotate. The model processes 132 features per frame (x, y, z coordinates + visibility) to distinguish 14 strike types. For stability, we implement a 10-frame smoothing buffer—if 7 of the last 10 frames are predicting a front kick, we commit to it as a kick. This prevents flipping between similar motions like head hooks vs. body hooks. The same expert recordings were used to train the model as DTW's reference templates, and therefore they have comparable baselines.

We conducted ablation studies on four progressively improved models to evaluate our feature engineering [8].

- **Keypoints only:** Using the raw (x, y, z) coordinates and visibility scores of all 33 body landmarks extracted via MediaPipe Pose.[2]

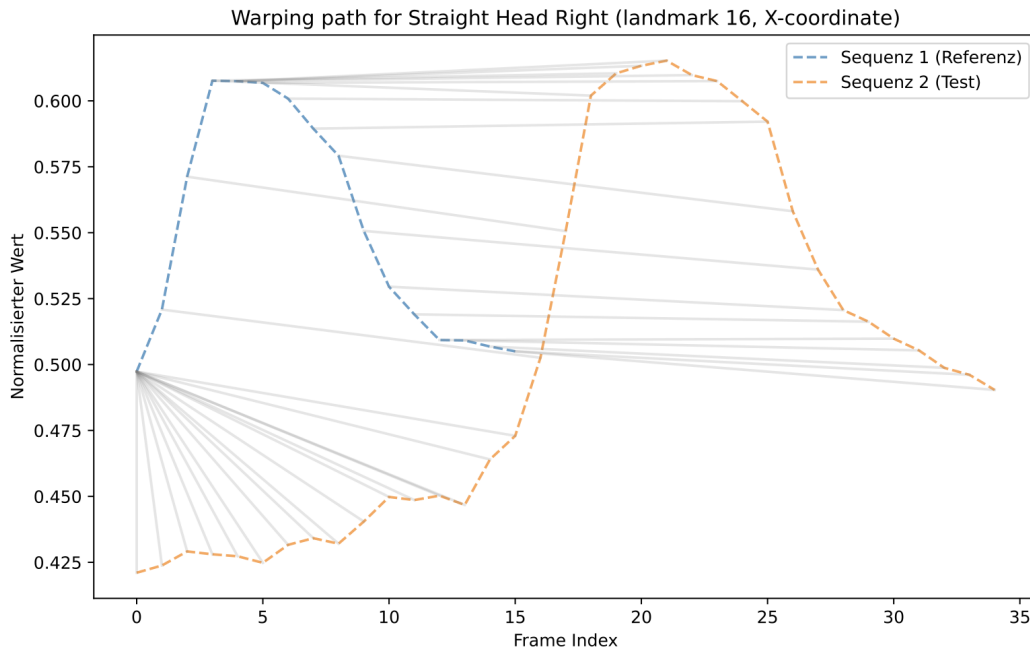


Figure 1: DTW Straight Head Right Example

- **Keypoints + Velocity:** Augmented the base feature set with temporal features such as wrist and ankle velocities.
- **Keypoints + Angles:** Included biomechanical features like the angles between upper and lower arms and legs to provide posture context.
- **Keypoints + Velocity + Angles + Angle Velocities:** Combined all previously mentioned features and added angle velocity to encode dynamic posture transitions.

Our results suggest that the combination of velocity and angle features with raw keypoints leads to more consistent classification results, especially for techniques with complex motion like kicks. Random Forest was chosen due to its robustness, non-linear decision capability, and success in human activity recognition tasks involving high-dimensional motion features.[9]

4 Development Process

The project began with a brainstorming session, and we all decided to work on Thai boxing as our topic of interest. We wanted to explore the feasibility of detecting punches and kicks using AI-based video analysis. Once we chose the topic, we established a shared workspace on Discord for coordination and communication. We divided the workload into data collection, data preprocessing, frame labeling, model construction and finally evaluation.

Initially, we searched for existing datasets for boxing and kicking actions. We quickly discovered that there are no free and appropriate datasets, which are Thai boxing technique-specific. That is why we decided to create our own dataset. This led us to approach our supervisor, Quentin and other Thai boxer of his dojo. With the utilization of three cameras with three angles, we were able to record multiple techniques, hence recording 14 different moves executed on both the left and right side. Each of the techniques was identified with a unique number for preprocessing.

We initially employed Roboflow [6] to annotate our video sequences to create our training and test sets. Our dataset for object Detection is publicly available on Roboflow.¹

We had originally intended to utilize the annotated data from Roboflow for object detection. Following some discussion with our supervisor Quentin, however, we decided that focusing on individual frames alone would not accurately capture the dynamic motion of Thai boxing techniques. Instead, we needed to come up with a way of considering frame sequences to more accurately depict the techniques. This understanding led us to change course towards a new approach.

¹<https://universe.roboflow.com/ai-lab-homm5/thai-boxing-object-detection>

Eventually, we used MediaPipe Pose [2] for feature extraction to obtain 33 skeletal keypoints from each video frame, which can be only applied to one person at a time. From the keypoints, we derived additional features, such as joint angles and velocities.

For the next phase, we designed our system around two methodologies that capture motion data before classifying the movements. For time reasons, we also choose the videos from only one angle. We explored two potential approaches:

- A template-matching approach using Dynamic Time Warping (DTW) [4].
- A supervised machine learning approach using Random Forest classification [5].

The extracted and preprocessed data was structured into time-series sequences suitable for analysis for the DTW-based approach. We created reference sequences from ideal executions and implemented the DTW algorithm to compare test sequences against these references, fine-tuning parameters such as window size and classification thresholds.

In parallel, we developed a Random Forest classifier, enriching the keypoint data with engineered features such as joint angles and velocities. A sliding window method was used to generate input sequences for training, and stratified sampling was applied to address class imbalance. To aid in interpretation and debugging, we also developed visualizations to display classification outcomes.

One of the most significant problems we faced in our project was the efficient handling of the complex Thai boxing moves. Each approach had its own set of problems that impacted overall performance.

The Random Forest Classifier had several problems. Occlusion problems were common where keypoints used to get obscured when body parts were occluded, leading to incomplete or error prone data. Moreover, using single static camera angles further made the problem more challenging, as rear-leg kicks were more observable than front-leg strikes, leading to inconsistent observability. Also, the dataset for the chosen camera angle was relatively small, making it hard for the model to generalize well. The inclusion of features like velocities and joint angles did enhance performance, but the benefits were small and short of really addressing the built-in issues.

The Dynamic Time Warping (DTW) method was most frustrating, failing to generalize properly. It was overly dependent on reference sequences, which is why this method is poorly capable of handling variability in Thai boxing movements. Sensitivity of camera angle and position of the fighter created significant disparities and stance, position and execution differences between different fighters prevented exact technique matching. Furthermore, differences in step size and speed of execution between input and reference sequences reduced the DTW accuracy even further.

5 Teamwork and Contribution Evaluation

We used Discord as our main communication and coordination platform. This allowed us to stay in sync through frequent voice calls, shared folders, and discussion threads. We scheduled weekly check-ins to track progress and troubleshoot as a group. During the process of data creation, every member participated actively in creating the videos and tagging the data. As the project progressed, we strategically assigned duties to increase its efficiency and avoid redundancy

Individual Contributions:

- Nils Kovacic: Took charge of DTW implementation and threshold tuning.
- Maximilian Laue: Managed MediaPipe integration and DTW implementation.
- Ebenhaezer Aubrey Sopacua: Focused on labeling and video segmentation.
- Mithusan Naguleswaran: Led the implementation of the Random Forest model and handled features.
- Tim Duc Minh: Coordinated documentation and evaluation.

Our collaboration was effective overall. We divided work clearly and supported each other during bottlenecks, especially around the preprocessing phase and the testing phase. While we initially struggled with different coding styles, we aligned our structure by the mid-point and documented shared functions more rigorously. For example for DTW, we combined different implementations and discussed our findings thoroughly via Discord meetings. The workload was overall fairly distributed and everyone delivered on their responsibilities. Especially in times of exams and stressful phases during the semester, the team spirit and communication were strong throughout the project. That is why everyone should receive the same grade.

6 Testing and Evaluation

To measure how well our system works, we tested both the Dynamic Time Warping (DTW) [4] and the Random Forest [5] methods on common evaluation metrics like accuracy and F1-scores. We used our manually labeled dataset, where four people were recorded and chose references of one person for testing the Random Forest Classifier and for DTW. The other three person were used for training. Specifically for DTW, we applied a threshold-based classification system and tested its performance in different scenarios, including variations in speed, technique angles and individual fighting styles.

Both approaches faced several common challenges. MediaPipe's restriction to single-person limited its applicability for sparring scenarios or multiple participants. Additionally, persistent issues such as the confusion of left and right and the keypoint occlusion affected the robustness of both models.

6.1 Random Forest Classifier

We can see that augmenting the dataset with body angles and velocity enhances model performance. The best model variant is the Random Forest Classifier with keypoints, velocity and angles as features to encode dynamic posture transitions.

The Random Forest classifier demonstrated moderate accuracy overall and its performance varied significantly across different technique categories (Figure 2). Techniques involving dynamic, full-body movements, were generally recognized with higher accuracy. In contrast, compact, posture-dependent motions proved to be more challenging to classify accurately. The differences in performance can be attributed to several factors. Firstly, the relatively small size of our dataset contributed to overfitting, particularly for less represented techniques, which limited the model's ability to generalize effectively. Secondly, the frequent occlusion of key body parts, such as arms or legs obscured by the body or punching bag, led to incomplete or noisy keypoint data and eventually to a complicated classification. Lastly, the fixed camera angle presented ambiguity when distinguishing between left- and right-sided techniques, hindering the model's ability to differentiate mirrored movements effectively.

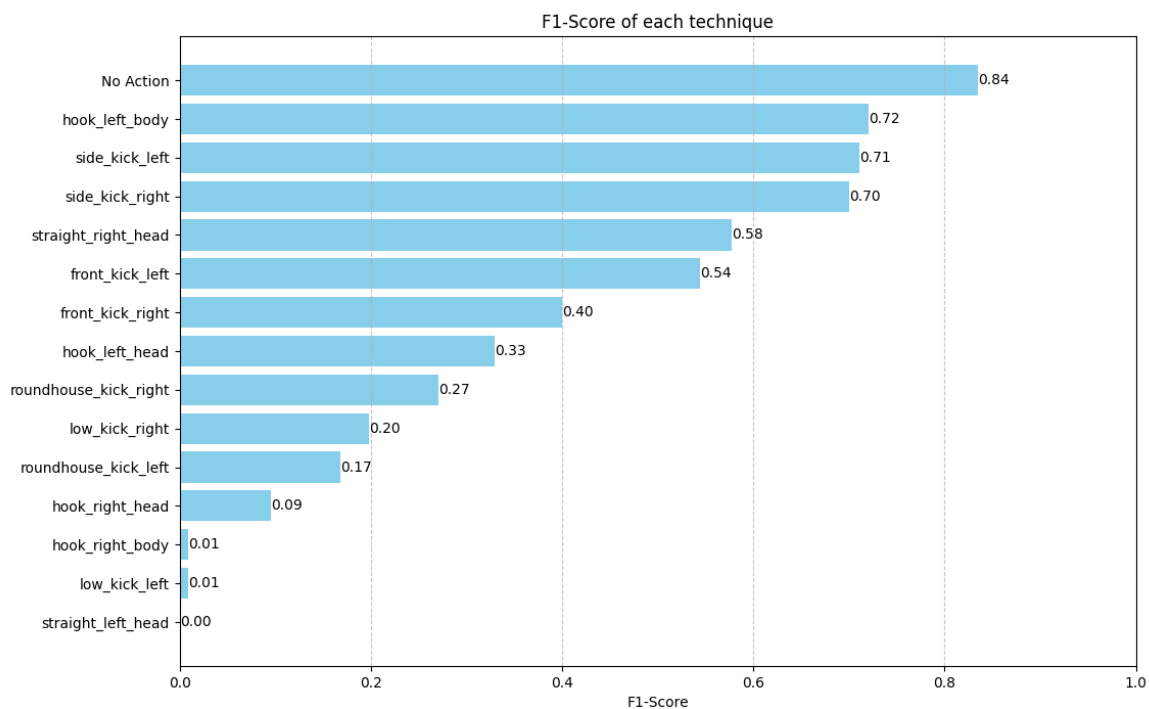


Figure 2: F1 Score for Each Techniques Random Forest

6.2 DTW

The Dynamic time Warping (DTW) method struggled to perform effectively on unseen test videos. Its reliance on direct comparisons with predefined reference sequences made DTW highly sensitive to even minor deviations in execution, such as variations in stance, motion speed or limb positioning. As a result, it often failed to recognize valid techniques when performed by individuals other than the reference fighter. This strong dependence on the specific execution style of the reference person highlights DTW's limited robustness in scenarios, where each fighter naturally performs the same technique with subtle differences. However, when applied to sequences from the original reference person, DTW delivered better results, which emphasized its inability to generalize across different fighters.

7 Conclusion and Reflection

In this project, we successfully developed a movement classification system for Thai boxing techniques by using a combination of MediaPipe for pose estimation and machine learning models for classification. Our approach demonstrated that it is possible to detect Thai boxing techniques with AI-driven methods based on skeletal keypoint data, which addresses the primary research question (RQ1). One of the key achievements was demonstrating that Zero/Few-Shot Learning can be used to detect fight movements in Thai boxing (Sub-RQ 1.1). Although Dynamic Time Warping (DTW) sounded as a good solution for our problem, it struggled with real-world variability and did not generalize well. In contrast, the Random Forest Classifier turned out to be a more reliable and flexible approach, especially when leveraging keypoints, velocity and angles as features. This enhancement confirmed that explicitly adding velocity and angle information improved the model performance (Sub-RQ 1.4). However, despite these improvements, the system faced some limitations. While feature extraction proved beneficial, it could not fully compensate for issues like occluded keypoints or limited spatial coverage. Additionally, the system could only effectively detect movements of a single person because MediaPipe struggled to detect two people in a fighting scenario (Sub-RQ 1.3). Furthermore, the results indicated that not all techniques could be equally detected from a single perspective (see Figure 2). Due to the inherent challenges posed by hidden body parts and varying angles, some techniques performed worse than others (Sub-RQ 1.5).

Overall, our project demonstrated the feasibility of detecting Thai boxing movements through AI-driven methods while also highlighting key limitations and areas for future improvement.

Throughout this project, we gained valuable insights into the challenges and complexities of detecting Thai boxing techniques via AI methods. One of the most important lessons was to understand that failure is an inherent part of research. Facing challenges often requires reevaluating initial approaches and being open to new methodologies. This realization highlights the importance of seeking guidance from our supervisor more proactively. We also learned that researching is not solely about achieving the best results but about exploring, understanding, and refining approaches even when outcomes are unexpected or failing. This allowed us to embrace setbacks as learning opportunities and refine our methods based on observed limitations.

Moreover, the project demonstrated that traditional object detection or keypoint detection on single frames is not well-suited for detecting dynamic and complex movements in Thai boxing. Because these techniques fail to capture the sequential and temporal nature of fighting techniques, which emphasizes the need for methods that consider sequences instead of single frames. Another critical takeaway was the importance of dataset diversity and camera viewpoint variety. The challenges we faced like the left and right ambiguity and the occlusion could be due to the limitations in our dataset and the limiting camera angles we used, as we selected videos from only one angle due to time constraints. Our experience with Dynamic Time Warping (DTW) further emphasized the weaknesses of template-matching methods in dynamic scenarios. While DTW can effectively align similar sequences, it struggles to handle the natural variability and deviations that occur in real-world combat scenarios. This insight reinforced the importance of using more adaptable and learning-based methods for complex action recognition tasks. Overall, this project taught us that flexibility, iterative refinement, and critical evaluation are essential when researching in AI.

While our project successfully demonstrated the feasibility of detecting Thai boxing techniques, there is a lot of improvement that could significantly enhance the system's performance and robustness in future work:

- 1. Expanding the Dataset: Increasing the diversity of the dataset by including more fighters, varied environments, and different fighting styles would help the model generalize better. A more comprehensive dataset would mitigate issues like the left and right ambiguity, leading to improved performance in real-world scenarios.
- 2. Integrating Multiple Camera Angles: Using multiple camera angles simultaneously could address keypoint loss due to occlusion and improve the classification of techniques that are mirrored or partially hidden. Multi-angle perspectives would enable the model to capture movements more accurately even when body parts are obscured from a single viewpoint.
- 3. Leveraging Deep Learning Models: Exploring deep learning models, tailored for action recognition, like Transformer-based approaches, could handle the dynamics and variability of Thai boxing techniques better than traditional classifiers. These models could learn complex motion patterns and generalize across fighters and scenarios better. [10], [11]

- 4. Addressing MediaPipe's Single-Person Limitation: To make the system suitable for more complex fighting scenarios, it's essential to overcome MediaPipe's single-person limitation. Potential solutions include integrating a multi-person pose estimation framework or implementing a preprocessing scheme to isolate individuals before processing. This would enable the system to handle interactions between multiple fighters more effectively.
- 5. Using a 3D Camera Perspective and AI for Enhanced Feature Extraction: Utilizing a 3D camera perspective to model fighters in three dimensions would offer more precise feature extraction by capturing depth and angles more accurately. AI-driven optimization could also be employed for dynamic threshold adjustments and fine-tuning, ensuring robust and adaptive performance in various conditions.

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